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Trigonometric Ratio and Identites

JEE Main Mathematics · Previous Year Questions with Answer Key

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MCQ (Single Correct Answer)	64 questions
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MCQ (Single Correct Answer) 64 questions

Q1. JEE Main 2026 (Online) 5th April Morning Shift

Let $\tan A, \tan B$, where $A, B \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, be the roots of the quadratic equation $x^2 - 2x - 5 = 0$. Then $20\sin^2\left(\frac{A+B}{2}\right)$ is equal to:

(A) $10 + \sqrt{10}$ (B) $10 - 2\sqrt{10}$ (C) 10-3 $\sqrt{10}$ (D) $10 - \sqrt{10}$

Open solution

Q2. JEE Main 2026 (Online) 2nd April Evening Shift

Let $P = \{\theta \in [0, 4\pi] : \tan^2\theta \neq 1\}$ and $S = \{a \in \mathbb{Z} : 2(\cos^8\theta - \sin^8\theta)\sec 2\theta = a^2, \theta \in P\}$. Then $n(S)$ is:

(A) 0 (B) 1 (C) 2 (D) 3

Open solution

Q3. JEE Main 2026 (Online) 2nd April Morning Shift

If $\sin\left(\frac{\pi}{18}\right)\sin\left(\frac{5\pi}{18}\right)\sin\left(\frac{7\pi}{18}\right) = K$, then the value of $\sin\left(\frac{10K\pi}{3}\right)$ is:

(A) $\frac{\sqrt{3}+1}{2\sqrt{2}}$ (B) $\frac{\sqrt{3}-1}{\sqrt{2}}$ (C) $\frac{\sqrt{3}}{2}$ (D) $\frac{1}{2}$

Open solution

Q4. JEE Main 2026 (Online) 28th January Morning Shift

If $\frac{\tan(A-B)}{\tan A} + \frac{\sin^2 C}{\sin^2 A} = 1,\ A, B, C \in \left(0, \frac{\pi}{2}\right)$, then

(A) $\tan A, \tan C, \tan B$ are in A.P. (B) $\tan A, \tan C, \tan B$ are in G.P.

(C) $\tan A, \tan B, \tan C$ are in G.P. (D) $\tan A, \tan B, \tan C$ are in A.P.

Open solution

Q5. JEE Main 2026 (Online) 24th January Morning Shift

The value of $\frac{\sqrt{3}\,\text{cosec}\,20^\circ - \sec 20^\circ}{\cos 20^\circ \cos 40^\circ \cos 60^\circ \cos 80^\circ}$ is equal to

(A) 32 (B) 64 (C) 12 (D) 16

Open solution

Q6. JEE Main 2026 (Online) 24th January Morning Shift

If $\cot x = \frac{5}{12}$ for some $x \in \left(\pi, \frac{3\pi}{2}\right)$, then $\sin 7x\left(\cos\frac{13x}{2} + \sin\frac{13x}{2}\right) + \cos 7x\left(\cos\frac{13x}{2} - \sin\frac{13x}{2}\right)$ is equal to

(A) $\frac{5}{\sqrt{13}}$ (B) $\frac{6}{\sqrt{26}}$ (C) $\frac{4}{\sqrt{26}}$ (D) $\frac{1}{\sqrt{13}}$

Open solution

Q7. JEE Main 2026 (Online) 23rd January Evening Shift

Let $\frac{\pi}{2} < \theta < \pi$ and $\cot\theta = -\frac{1}{2\sqrt{2}}$. Then the value of

$\sin\left(\frac{15\theta}{2}\right)(\cos 8\theta + \sin 8\theta) + \cos\left(\frac{15\theta}{2}\right)(\cos 8\theta - \sin 8\theta)$

is equal to :

(A) $\frac{\sqrt{2}-1}{\sqrt{3}}$ (B) $\frac{\sqrt{2}}{\sqrt{3}}$ (C) $\frac{1-\sqrt{2}}{\sqrt{3}}$ (D) $-\frac{\sqrt{2}}{\sqrt{3}}$

Open solution

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Q8. JEE Main 2026 (Online) 21st January Morning Shift

The value of $\operatorname{cosec} 10^\circ - \sqrt{3}\sec 10^\circ$ is equal to :

(A) 2 (B) 6 (C) 8 (D) 4

Open solution

Q9. JEE Main 2025 (Online) 7th April Morning Shift

If for $\theta \in \left[-\frac{\pi}{3}, 0\right]$, the points $(x, y) = \left(3\tan\left(\theta + \frac{\pi}{3}\right), 2\tan\left(\theta + \frac{\pi}{6}\right)\right)$ lie on $xy + \alpha x + \beta y + \gamma = 0$, then $\alpha^2 + \beta^2 + \gamma^2$ is equal to :

(A) 75 (B) 96 (C) 80 (D) 72

Open solution

Q10. JEE Main 2025 (Online) 4th April Morning Shift

If $10\sin^4\theta + 15\cos^4\theta = 6$, then the value of $\frac{27\operatorname{cosec}^6\theta + 8\sec^6\theta}{16\sec^8\theta}$ is

(A) $\frac{2}{5}$ (B) $\frac{3}{5}$ (C) $\frac{1}{5}$ (D) $\frac{3}{4}$

Open solution

Q11. JEE Main 2025 (Online) 29th January Evening Shift

If $\sin x + \sin^2 x = 1, x \in \left(0, \frac{\pi}{2}\right)$, then

$(\cos^{12} x + \tan^{12} x) + 3(\cos^{10} x + \tan^{10} x + \cos^8 x + \tan^8 x) + (\cos^6 x + \tan^6 x)$ is equal to:

(A) 3 (B) 4 (C) 2 (D) 1

Open solution

Q12. JEE Main 2025 (Online) 28th January Evening Shift

If $\sum_{r=1}^{13}\left\{\frac{1}{\sin\left(\frac{\pi}{4}+(r-1)\frac{\pi}{6}\right)\sin\left(\frac{\pi}{4}+\frac{r\pi}{6}\right)}\right\} = a\sqrt{3} + b, a, b \in Z$, then $a^2 + b^2$ is equal to :

(A) 10 (B) 4 (C) 8 (D) 2

Open solution

Q13. JEE Main 2025 (Online) 23rd January Evening Shift

Let the range of the function $f(x) = 6 + 16\cos x \cdot \cos\left(\frac{\pi}{3} - x\right) \cdot \cos\left(\frac{\pi}{3} + x\right) \cdot \sin 3x \cdot \cos 6x, x \in \mathbf{R}$ be $[\alpha, \beta]$. Then the distance of the point (α, β) from the line $3x + 4y + 12 = 0$ is :

(A) 11 (B) 10 (C) 8 (D) 9

Open solution

Q14. JEE Main 2025 (Online) 23rd January Morning Shift

The value of $(\sin 70^\circ)(\cot 10^\circ \cot 70^\circ - 1)$ is

(A) 0 (B) 2/3 (C) 1 (D) 3/2

Open solution

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Q15. JEE Main 2024 (Online) 8th April Evening Shift

If the value of $\frac{3\cos 36^\circ+5\sin 18^\circ}{5\cos 36^\circ-3\sin 18^\circ}$ is $\frac{a\sqrt{5}-b}{c}$, where a, b, c are natural numbers and $\gcd(a, c) = 1$, then $a + b + c$ is equal to :

(A) 54 (B) 52 (C) 50 (D) 40

Open solution

Q16. JEE Main 2024 (Online) 8th April Morning Shift

If $\sin x = -\frac{3}{5}$, where $\pi < x < \frac{3\pi}{2}$, then $80\left(\tan^2 x - \cos x\right)$ is equal to

(A) 109 (B) 108 (C) 19 (D) 18

Open solution

Q17. JEE Main 2024 (Online) 1st February Morning Shift

If $\tan A = \frac{1}{\sqrt{x(x^2+x+1)}}$, $\tan B = \frac{\sqrt{x}}{\sqrt{x^2+x+1}}$ and

$\tan C = \left(x^{-3} + x^{-2} + x^{-1}\right)^{1/2}, 0 < A, B, C < \frac{\pi}{2}$, then $A + B$ is equal to :

(A) C (B) $\pi - C$ (C) $2\pi - C$ (D) $\frac{\pi}{2} - C$

Open solution

Q18. JEE Main 2024 (Online) 31st January Evening Shift

The number of solutions, of the equation $e^{\sin x} - 2e^{-\sin x} = 2$, is :

(A) 0 (B) 1 (C) 2 (D) more than 2

Open solution

Q19. JEE Main 2024 (Online) 30th January Evening Shift

For $\alpha, \beta \in (0, \pi/2)$, let $3\sin(\alpha + \beta) = 2\sin(\alpha - \beta)$ and a real number k be such that $\tan\alpha = k\tan\beta$. Then, the value of k is equal to

(A) 5 (B) $-2/3$ (C) -5 (D) 2/3

Open solution

Q20. JEE Main 2023 (Online) 10th April Morning Shift

$96\cos\frac{\pi}{33}\cos\frac{2\pi}{33}\cos\frac{4\pi}{33}\cos\frac{8\pi}{33}\cos\frac{16\pi}{33}$ is equal to :

(A) 4 (B) 2 (C) 1 (D) 3

Open solution

Q21. JEE Main 2023 (Online) 8th April Evening Shift

The value of $36\left(4\cos^2 9^\circ - 1\right)\left(4\cos^2 27^\circ - 1\right)\left(4\cos^2 81^\circ - 1\right)\left(4\cos^2 243^\circ - 1\right)$ is :

(A) 18 (B) 36 (C) 54 (D) 27

Open solution

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Q22. JEE Main 2023 (Online) 30th January Morning Shift

If $\tan 15^\circ + \frac{1}{\tan 75^\circ} + \frac{1}{\tan 105^\circ} + \tan 195^\circ = 2a$, then the value of $\left(a + \frac{1}{a}\right)$ is :

(A) $5 - \frac{3}{2}\sqrt{3}$ (B) $4 - 2\sqrt{3}$ (C) 2 (D) 4

Open solution

Q23. JEE Main 2023 (Online) 29th January Evening Shift

The set of all values of λ for which the equation $\cos^2 2x - 2\sin^4 x - 2\cos^2 x = \lambda$ has a real solution x, is :

(A) $[-2, -1]$ (B) $\left[-\frac{3}{2}, -1\right]$

(C) $\left[-2, -\frac{3}{2}\right]$ (D) $\left[-1, -\frac{1}{2}\right]$

Open solution

Q24. JEE Main 2023 (Online) 29th January Morning Shift

Let $f(\theta) = 3\left(\sin^4\left(\frac{3\pi}{2} - \theta\right) + \sin^4(3\pi + \theta)\right) - 2(1 - \sin^2 2\theta)$ and $S = \left\{\theta \in [0, \pi] : f'(\theta) = -\frac{\sqrt{3}}{2}\right\}$. If $4\beta = \sum_{\theta \in S} \theta$, then $f(\beta)$ is equal to

(A) $\frac{9}{8}$ (B) $\frac{3}{2}$ (C) $\frac{5}{4}$ (D) $\frac{11}{8}$

Open solution

Q25. JEE Main 2022 (Online) 25th July Evening Shift

$2\sin\left(\frac{\pi}{22}\right)\sin\left(\frac{3\pi}{22}\right)\sin\left(\frac{5\pi}{22}\right)\sin\left(\frac{7\pi}{22}\right)\sin\left(\frac{9\pi}{22}\right)$ is equal to :

(A) $\frac{3}{16}$ (B) $\frac{1}{16}$ (C) $\frac{1}{32}$ (D) $\frac{9}{32}$

Open solution

Q26. JEE Main 2022 (Online) 28th June Evening Shift

If $\cot\alpha$ = 1 and $\sec\beta = -\frac{5}{3}$, where $\pi < \alpha < \frac{3\pi}{2}$ and $\frac{\pi}{2} < \beta < \pi$, then the value of $\tan(\alpha + \beta)$ and the quadrant in which $\alpha + \beta$ lies, respectively are :

(A) $-\frac{1}{7}$ and IVth quadrant (B) 7 and Ist quadrant

(C) -7 and IVth quadrant (D) $\frac{1}{7}$ and Ist quadrant

Open solution

Q27. JEE Main 2022 (Online) 27th June Evening Shift

$\alpha = \sin 36^\circ$ is a root of which of the following equation?

(A) $16x^4 - 10x^2 - 5 = 0$ (B) $16x^4 + 20x^2 - 5 = 0$

(C) $16x^4 - 20x^2 + 5 = 0$ (D) $4x^4 - 10x^2 + 5 = 0$

Open solution

Q28. JEE Main 2022 (Online) 27th June Morning Shift

The value of $\cos\left(\frac{2\pi}{7}\right) + \cos\left(\frac{4\pi}{7}\right) + \cos\left(\frac{6\pi}{7}\right)$ is equal to :

(A) -1 (B) $-\frac{1}{2}$ (C) $-\frac{1}{3}$ (D) $-\frac{1}{4}$

Open solution

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Q29. JEE Main 2022 (Online) 26th June Evening Shift

$16\sin(20^\circ)\sin(40^\circ)\sin(80^\circ)$ is equal to :

(A) $\sqrt{3}$ (B) $2\sqrt{3}$ (C) 3 (D) $4\sqrt{3}$

Open solution

Q30. JEE Main 2022 (Online) 25th June Evening Shift

The value of $2\sin(12^\circ) - \sin(72^\circ)$ is :

(A) $\frac{\sqrt{5}(1-\sqrt{3})}{4}$ (B) $\frac{1-\sqrt{5}}{8}$

(C) $\frac{\sqrt{3}(1-\sqrt{5})}{2}$ (D) $\frac{\sqrt{3}(1-\sqrt{5})}{4}$

Open solution

Q31. JEE Main 2021 (Online) 26th August Evening Shift

The value of

$2\sin\left(\frac{\pi}{8}\right)\sin\left(\frac{2\pi}{8}\right)\sin\left(\frac{3\pi}{8}\right)\sin\left(\frac{5\pi}{8}\right)\sin\left(\frac{6\pi}{8}\right)\sin\left(\frac{7\pi}{8}\right)$ is :

(A) $\frac{1}{4\sqrt{2}}$ (B) $\frac{1}{4}$ (C) $\frac{1}{8}$ (D) $\frac{1}{8\sqrt{2}}$

Open solution

Q32. JEE Main 2021 (Online) 27th July Evening Shift

If $\tan\left(\frac{\pi}{9}\right), x, \tan\left(\frac{7\pi}{18}\right)$ are in arithmetic progression and $\tan\left(\frac{\pi}{9}\right), y, \tan\left(\frac{5\pi}{18}\right)$ are also in arithmetic progression, then $|x-2y|$ is equal to :

(A) 4 (B) 3 (C) 0 (D) 1

Open solution

Q33. JEE Main 2021 (Online) 27th July Morning Shift

If $\sin\theta + \cos\theta = \frac{1}{2}$, then 16(sin(2$\theta$) + cos(4$\theta$) + sin(6$\theta$)) is equal to :

(A) 23 (B) -27 (C) -23 (D) 27

Open solution

Q34. JEE Main 2021 (Online) 25th July Evening Shift

The value of $\cot\frac{\pi}{24}$ is :

(A) $\sqrt{2}+\sqrt{3}+2-\sqrt{6}$ (B) $\sqrt{2}+\sqrt{3}+2+\sqrt{6}$

(C) $\sqrt{2}-\sqrt{3}-2+\sqrt{6}$ (D) $3\sqrt{2}-\sqrt{3}-\sqrt{6}$

Open solution

Q35. JEE Main 2021 (Online) 18th March Evening Shift

If $15\sin^4\alpha + 10\cos^4\alpha = 6$, for some $\alpha \in R$, then the value of

$27\sec^6\alpha + 8\operatorname{cosec}^6\alpha$ is equal to :

(A) 500 (B) 400 (C) 250 (D) 350

Open solution

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Q36. JEE Main 2021 (Online) 16th March Morning Shift

If for x $\in \left(0, \frac{\pi}{2}\right)$, $\log_{10}\sin x + \log_{10}\cos x = -1$ and $\log_{10}(\sin x + \cos x) = \frac{1}{2}(\log_{10} n - 1)$, n > 0, then the value of n is equal to :

(A) 16 (B) 9 (C) 12 (D) 20

Open solution

Q37. JEE Main 2021 (Online) 25th February Evening Shift

If 0 < x, y < π and $\cos x + \cos y - \cos(x + y) = \frac{3}{2}$, then $\sin x + \cos y$ is equal to :

(A) $\frac{1+\sqrt{3}}{2}$ (B) $\frac{1}{2}$ (C) $\frac{\sqrt{3}}{2}$ (D) $\frac{1-\sqrt{3}}{2}$

Open solution

Q38. JEE Main 2021 (Online) 24th February Morning Shift

If $e^{(\cos^2 x+\cos^4 x+\cos^6 x+\ldots\infty)\log_e 2}$ satisfies the equation $t^2 - 9t + 8 = 0$, then the value of $\frac{2\sin x}{\sin x+\sqrt{3}\cos x}\left(0 < x < \frac{\pi}{2}\right)$ is :

(A) $\sqrt{3}$ (B) $\frac{3}{2}$ (C) $2\sqrt{3}$ (D) $\frac{1}{2}$

Open solution

Q39. JEE Main 2020 (Online) 5th September Evening Slot

If $L = \sin^2\left(\frac{\pi}{16}\right) - \sin^2\left(\frac{\pi}{8}\right)$ and $M = \cos^2\left(\frac{\pi}{16}\right) - \sin^2\left(\frac{\pi}{8}\right)$, then :

(A) $L = -\frac{1}{2\sqrt{2}} + \frac{1}{2}\cos\frac{\pi}{8}$ (B) $M = \frac{1}{2\sqrt{2}} + \frac{1}{2}\cos\frac{\pi}{8}$

(C) $M = \frac{1}{4\sqrt{2}} + \frac{1}{4}\cos\frac{\pi}{8}$ (D) $L = \frac{1}{4\sqrt{2}} - \frac{1}{4}\cos\frac{\pi}{8}$

Open solution

Q40. JEE Main 2020 (Online) 2nd September Evening Slot

If the equation $\cos^4\theta + \sin^4\theta + \lambda = 0$ has real solutions for θ, then λ lies in the interval :

(A) $\left[-\frac{3}{2}, -\frac{5}{4}\right]$ (B) $\left(-\frac{1}{2}, -\frac{1}{4}\right]$

(C) $\left(-\frac{5}{4}, -1\right]$ (D) $\left[-1, -\frac{1}{2}\right]$

Open solution

Q41. JEE Main 2020 (Online) 9th January Evening Slot

If $x = \sum_{n=0}^{\infty}(-1)^n \tan^{2n}\theta$ and $y = \sum_{n=0}^{\infty}\cos^{2n}\theta$

for 0 < θ < $\frac{\pi}{4}$, then :

(A) x(1 + y) = 1 (B) y(1 − x) = 1 (C) y(1 + x) = 1 (D) x(1 − y) = 1

Open solution

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Q42. JEE Main 2020 (Online) 9th January Morning Slot

The value of
$\cos^3\left(\frac{\pi}{8}\right)\cos\left(\frac{3\pi}{8}\right)+\sin^3\left(\frac{\pi}{8}\right)\sin\left(\frac{3\pi}{8}\right)$
is :

(A) $\frac{1}{\sqrt{2}}$ (B) $\frac{1}{2}$ (C) $\frac{1}{4}$ (D) $\frac{1}{2\sqrt{2}}$

Open solution

Q43. JEE Main 2019 (Online) 12th April Morning Slot

The equation y = sinx sin (x + 2) − $\sin^2$ (x + 1) represents a straight line lying in :

(A) first, second and fourth quadrants
(B) first, third and fourth quadrants
(C) second and third quadrants only
(D) third and fourth quadrants only

Open solution

Q44. JEE Main 2019 (Online) 9th April Evening Slot

The value of sin 10° sin30° sin50° sin70° is :-

(A) $\frac{1}{36}$ (B) $\frac{1}{16}$ (C) $\frac{1}{32}$ (D) $\frac{1}{18}$

Open solution

Q45. JEE Main 2019 (Online) 9th April Morning Slot

The value of $\cos^2 10° - \cos 10° \cos 50° + \cos^2 50°$ is

(A) $\frac{3}{2} + \cos 20^o$ (B) $\frac{3}{4}$ (C) $\frac{3}{2}(1 + \cos 20^o)$ (D) $\frac{3}{2}$

Open solution

Q46. JEE Main 2019 (Online) 8th April Morning Slot

If $\cos(\alpha + \beta) = 3/5$, $\sin(\alpha - \beta) = 5/13$ and $0 < \alpha, \beta < \frac{\pi}{4}$, then $\tan(2\alpha)$ is equal to :

(A) 21/16 (B) 63/52 (C) 33/52 (D) 63/16

Open solution

Q47. JEE Main 2019 (Online) 12th January Morning Slot

The maximum value of $3\cos\theta + 5\sin\left(\theta - \frac{\pi}{6}\right)$ for any real value of θ is :

(A) $\sqrt{34}$ (B) $\sqrt{31}$ (C) $\sqrt{19}$ (D) $\frac{\sqrt{79}}{2}$

Open solution

Q48. JEE Main 2019 (Online) 10th January Evening Slot

The value of $\cos\frac{\pi}{2^2}.\cos\frac{\pi}{2^3}.....\cos\frac{\pi}{2^{10}}.\sin\frac{\pi}{2^{10}}$ is -

(A) $\frac{1}{256}$ (B) $\frac{1}{2}$ (C) $\frac{1}{1024}$ (D) $\frac{1}{512}$

Open solution

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Q49. JEE Main 2019 (Online) 9th January Morning Slot

For any $\theta \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$, the expression

$3(\cos\theta - \sin\theta)^4 + 6(\sin\theta + \cos\theta)^2 + 4\sin^6\theta$

equals :

(A) $13 - 4\cos^2\theta + 6\sin^2\theta\cos^2\theta$

(B) $13 - 4\cos^6\theta$

(C) $13 - 4\cos^2\theta + 6\cos^4\theta$

(D) $13 - 4\cos^4\theta + 2\sin^2\theta\cos^2\theta$

Open solution

Q50. JEE Main 2017 (Offline)

If $5\left(\tan^2 x - \cos^2 x\right) = 2\cos 2x + 9$,

then the value of $\cos 4x$ is :

(A) $\frac{1}{3}$ (B) $\frac{2}{9}$ (C) $-\frac{7}{9}$ (D) $-\frac{3}{5}$

Open solution

Q51. JEE Main 2016 (Online) 9th April Morning Slot

If m and M are the minimum and the maximum values of

$4 + \frac{1}{2}\sin^2 2x - 2\cos^4 x$, $x \in \mathbf{R}$, then $M - m$ is equal to :

(A) $\frac{15}{4}$ (B) $\frac{9}{4}$ (C) $\frac{7}{4}$ (D) $\frac{1}{4}$

Open solution

Q52. JEE Main 2015 (Online) 11th April Morning Slot

If $\cos\alpha + \cos\beta = \frac{3}{2}$ and $\sin\alpha + \sin\beta = \frac{1}{2}$

and θ is the arithmetic mean of α and β,

then $\sin 2\theta + \cos 2\theta$ is equal to :

(A) $\frac{3}{5}$ (B) $\frac{4}{5}$ (C) $\frac{7}{5}$ (D) $\frac{8}{5}$

Open solution

Q53. JEE Main 2014 (Online) 11th April Morning Slot

If $2\cos\theta + \sin\theta = 1$ $\left(\theta \neq \frac{\pi}{2}\right)$
then $7\cos\theta + 6\sin\theta$ is equal to :

(A) $\frac{1}{2}$ (B) 2 (C) $\frac{11}{2}$ (D) $\frac{46}{5}$

Open solution

Q54. JEE Main 2014 (Online) 9th April Morning Slot

If $\operatorname{cosec}\theta = \frac{p+q}{p-q}$ $(p \neq q,\ p \neq 0)$, then $\left|\cot\left(\frac{\pi}{4} + \frac{\theta}{2}\right)\right|$ is equal to :

(A) pq (B) $\sqrt{\frac{p}{q}}$ (C) $\sqrt{\frac{q}{p}}$ (D) $\sqrt{pq}$

Open solution

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Q55. JEE Main 2014 (Offline)

Let $f_k(x) = \frac{1}{k}\left(\sin^k x + \cos^k x\right)$ where $x \in R$ and $k \geq 1$.
Then $f_4(x) - f_6(x)$ equals :

(A) $\frac{1}{4}$ (B) $\frac{1}{12}$ (C) $\frac{1}{6}$ (D) $\frac{1}{3}$

Open solution

Q56. JEE Main 2013 (Offline)

The expression $\frac{\tan A}{1-\cot A} + \frac{\cot A}{1-\tan A}$ can be written as:

(A) $\sin A \cos A + 1$ (B) $\sec A \cos ecA + 1$

(C) $\tan A + \cot A$ (D) $\sec A + \cos ecA$

Open solution

Q57. AIEEE 2012 (Online) 26th May

The value of $\cos 255^\circ + \sin 195^\circ$ is

(A) $\frac{\sqrt{3}-1}{2\sqrt{2}}$ (B) $\frac{\sqrt{3}-1}{\sqrt{2}}$ (C) $-\frac{\sqrt{3}-1}{\sqrt{2}}$ (D) $\frac{\sqrt{3}+1}{\sqrt{2}}$

Open solution

Q58. AIEEE 2012 (Online) 19th May

Suppose θ and $\phi\,(\neq 0)$ are such that $\sec(\theta+\phi)$, $\sec\theta$ and $\sec(\theta-\phi)$ are in A.P. If $\cos\theta = k\cos\left(\frac{\phi}{2}\right)$ for some k, then k is equal to

(A) $\pm\sqrt{2}$ (B) ± 1 (C) $\pm\frac{1}{\sqrt{2}}$ (D) ± 2

Open solution

Q59. AIEEE 2011

If $A = \sin^2 x + \cos^4 x$, then for all real x:

(A) $\frac{13}{16} \leq A \leq 1$ (B) $1 \leq A \leq 2$

(C) $\frac{3}{4} \leq A \leq \frac{13}{16}$ (D) $\frac{3}{4} \leq A \leq 1$

Open solution

Q60. AIEEE 2010

Let $\cos(\alpha+\beta) = \frac{4}{5}$ and $\sin(\alpha-\beta) = \frac{5}{13}$, where $0 \leq \alpha, \beta \leq \frac{\pi}{4}$.
Then $tan\,2\alpha$ =

(A) $\frac{56}{33}$ (B) $\frac{19}{12}$ (C) $\frac{20}{7}$ (D) $\frac{25}{16}$

Open solution

Q61. AIEEE 2009

Let **A** and **B** denote the statements

A: $\cos\alpha + \cos\beta + \cos\gamma = 0$

B: $\sin\alpha + \sin\beta + \sin\gamma = 0$

If $\cos(\beta-\gamma) + \cos(\gamma-\alpha) + \cos(\alpha-\beta) = -\frac{3}{2}$, then:

(A) **A** is false and **B** is true (B) both **A** and **B** are true

(C) both **A** and **B** are false (D) **A** is true and **B** is false

Open solution

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Q62. AIEEE 2006

If $0 < x < \pi$ and $\cos x + \sin x = \frac{1}{2}$, then $\tan x$ is :

(A) $\frac{\left(1-\sqrt{7}\right)}{4}$ (B) $\frac{\left(4-\sqrt{7}\right)}{3}$

(C) $-\frac{\left(4+\sqrt{7}\right)}{3}$ (D) $\frac{\left(1+\sqrt{7}\right)}{4}$

Open solution

Q63. AIEEE 2004

Let α, β be such that $\pi < \alpha - \beta < 3\pi$.
If $\sin\alpha + \sin\beta = -\frac{21}{65}$ and $\cos\alpha + \cos\beta = -\frac{27}{65}$ then the value of $\cos\frac{\alpha-\beta}{2}$:

(A) $\frac{-6}{65}$ (B) $\frac{3}{\sqrt{130}}$ (C) $\frac{6}{65}$ (D) $-\frac{3}{\sqrt{130}}$

Open solution

Q64. AIEEE 2004

If $u = \sqrt{a^2\cos^2\theta + b^2\sin^2\theta} + \sqrt{a^2\sin^2\theta + b^2\cos^2\theta}$

then the difference between the maximum and minimum values of u^2 is given by :

(A) $(a-b)^2$ (B) $2\sqrt{a^2+b^2}$ (C) $(a+b)^2$ (D) $2\left(a^2+b^2\right)$

Open solution

Numerical 9 questions

Q65. JEE Main 2026 (Online) 4th April Morning Shift

If $\mathrm{A} = \frac{\sin 3^\circ}{\cos 9^\circ} + \frac{\sin 9^\circ}{\cos 27^\circ} + \frac{\sin 27^\circ}{\cos 81^\circ}$ and $\mathrm{B} = \tan 81^\circ - \tan 3^\circ$, then $\frac{\mathrm{B}}{\mathrm{A}}$ is equal to ____ .

Open solution

Q66. JEE Main 2026 (Online) 4th April Morning Shift

Let $\overrightarrow{a_k} = (\tan\theta_k)\hat{i} + \hat{j}$ and $\overrightarrow{b_k} = \hat{i} - (\cot\theta_k)\hat{j}$, where $\theta_k = \frac{2^{k-1}\pi}{2^n+1}$, for some $n \in \mathbb{N}, n > 5$. Then the value of $\frac{\sum_{k=1}^{n}\left|\overrightarrow{a_k}\right|^2}{\sum_{k=1}^{n}\left|\overrightarrow{b_k}\right|^2}$ is

Open solution

Q67. JEE Main 2026 (Online) 22nd January Evening Shift

Let $\cos(\alpha+\beta) = -\frac{1}{10}$ and $\sin(\alpha-\beta) = \frac{3}{8}$, where $0 < \alpha < \frac{\pi}{3}$ and $0 < \beta < \frac{\pi}{4}$. If $\tan 2\alpha = \frac{3(1-r\sqrt{5})}{\sqrt{11}(s+\sqrt{5})}$, $r, s \in N$, then $r+s$ is equal to ____ .

Open solution

Q68. JEE Main 2026 (Online) 22nd January Morning Shift

If $\frac{\cos^2 48^\circ - \sin^2 12^\circ}{\sin^2 24^\circ - \sin^2 6^\circ} = \frac{\alpha+\beta\sqrt{5}}{2}$, where $\alpha, \beta \in \mathbb{N}$, then $\alpha+\beta$ is equal to ____

Open solution

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Q69. JEE Main 2024 (Online) 27th January Morning Shift

Let the set of all $a \in \mathbf{R}$ such that the equation $\cos 2x + a \sin x = 2a - 7$ has a solution be $[p, q]$ and $r = \tan 9^\circ - \tan 27^\circ - \frac{1}{\cot 63^\circ} + \tan 81^\circ$, then pqr is equal to _____.

Open solution

Q70. JEE Main 2023 (Online) 6th April Evening Shift

The value of $\tan 9^\circ - \tan 27^\circ - \tan 63^\circ + \tan 81^\circ$ is _____.

Open solution

Q71. JEE Main 2022 (Online) 26th June Morning Shift

If $\sin^2(10^\circ)\sin(20^\circ)\sin(40^\circ)\sin(50^\circ)\sin(70^\circ) = \alpha - \frac{1}{16}\sin(10^\circ)$, then $16 + \alpha^{-1}$ is equal to _____.

Open solution

Q72. JEE Main 2021 (Online) 26th February Morning Shift

The number of integral values of 'k' for which the equation $3\sin x + 4\cos x = k + 1$ has a solution, k∈R is _____.

Open solution

Q73. JEE Main 2020 (Online) 8th January Evening Slot

If $\frac{\sqrt{2}\sin\alpha}{\sqrt{1+\cos 2\alpha}} = \frac{1}{7}$ and $\sqrt{\frac{1-\cos 2\beta}{2}} = \frac{1}{\sqrt{10}}$

$\alpha, \beta \in \left(0, \frac{\pi}{2}\right)$ then tan(α + 2β) is equal to ____.

Open solution

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Answer Key

MCQ (Single Correct Answer)

1. **C**	2. **A**	3. **A**	4. **B**	5. **B**
6. **D**	7. **C**	8. **D**	9. **A**	10. **A**
11. **C**	12. **C**	13. **A**	14. **C**	15. **B**
16. **A**	17. **A**	18. **A**	19. **C**	20. **D**
21. **B**	22. **D**	23. **B**	24. **C**	25. **B**
26. **A**	27. **C**	28. **B**	29. **B**	30. **D**
31. **C**	32. **C**	33. **C**	34. **B**	35. **C**
36. **C**	37. **A**	38. **D**	39. **B**	40. **D**
41. **B**	42. **D**	43. **D**	44. **B**	45. **B**
46. **D**	47. **C**	48. **D**	49. **B**	50. **C**
51. **B**	52. **C**	53. **B**	54. **C**	55. **B**
56. **B**	57. **C**	58. **A**	59. **D**	60. **A**
61. **B**	62. **C**	63. **D**	64. **A**	

Numerical

65. **2**	66. **3**	67. **20**	68. **4**	69. **48**
70. **4**	71. **80**	72. **11**	73. **1**	

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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